ExpNo 16 Experiment to understand EDA Quantitative and Qualitative Analysis

Program:

```
import seaborn as sns

import pandas as pd

import numpy as np

import matplotlib.pyplot as plt


%matplotlib inline


tips = pd.read_csv('tips_modified.csv')

tips.head()


# Distribution plots

sns.displot(tips.total_bill, kde=True)

plt.show()


sns.displot(tips.total_bill, kde=False)

plt.show()


# Joint plots

sns.jointplot(x=tips.tip, y=tips.total_bill)

plt.show()


sns.jointplot(x=tips.tip, y=tips.total_bill, kind="reg")
```

```
plt.show()

sns.jointplot(x=tips.tip, y=tips.total_bill, kind="hex")
plt.show()

# Pair plots
sns.pairplot(tips)
plt.show()

tips.time.value_counts()

sns.pairplot(tips, hue='time')
plt.show()

sns.pairplot(tips, hue='day')
plt.show()

# Heatmap
plt.figure(figsize=(8, 5))
sns.heatmap(tips.corr(numeric_only=True), annot=True)
plt.show()

# Box plots
plt.figure()
sns.boxplot(x=tips.total_bill)
```

```
plt.show()


plt.figure()
sns.boxplot(x=tips.tip)
plt.show()

# Count plots
plt.figure()
sns.countplot(x=tips.day)
plt.show()

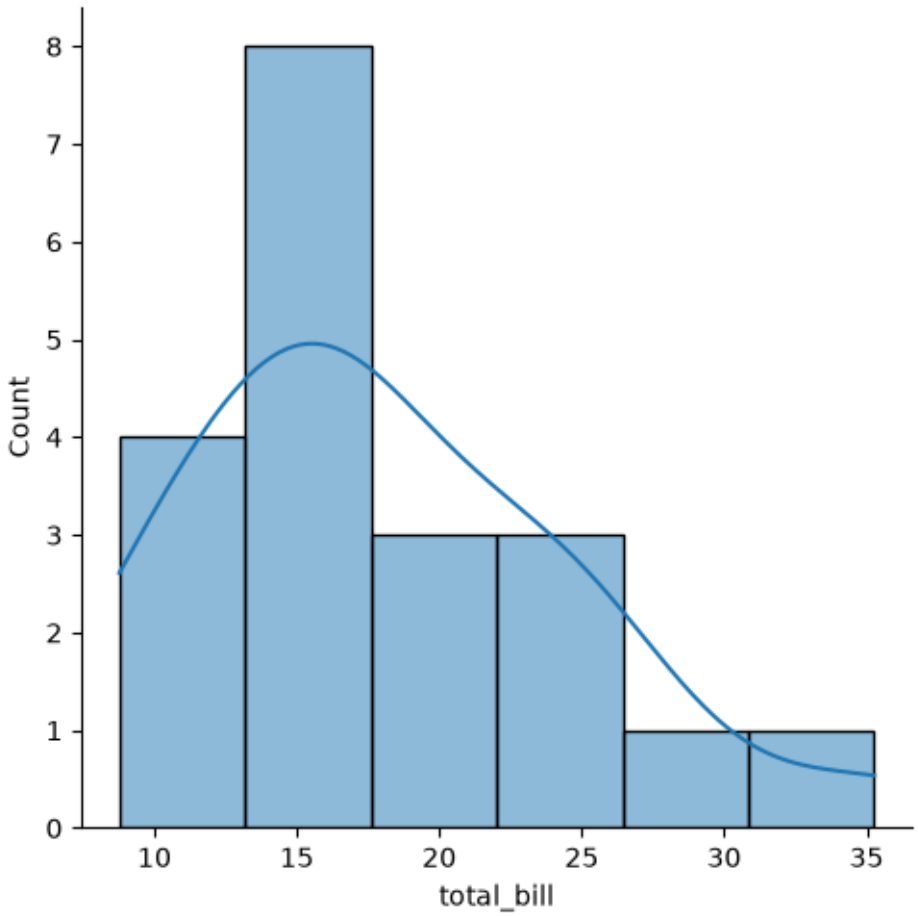
plt.figure()
sns.countplot(x=tips.sex)
plt.show()

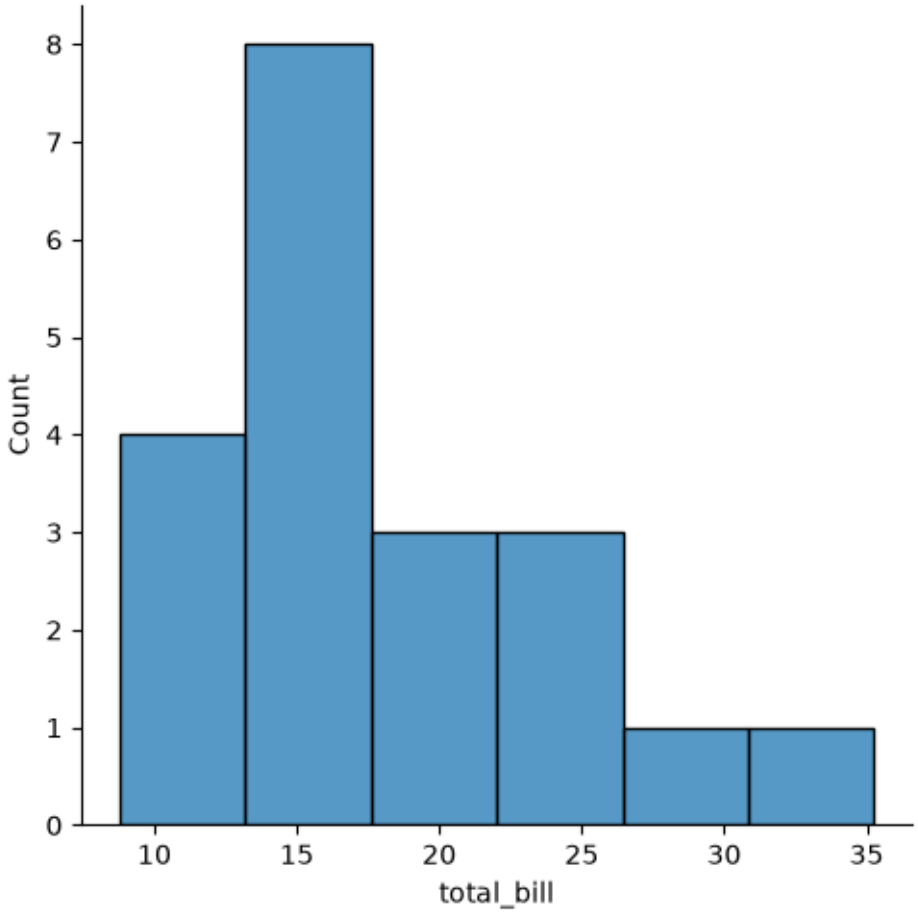
# Pie chart
plt.figure()
tips.sex.value_counts().plot(kind='pie', autopct='%1.1f%%')
plt.show()

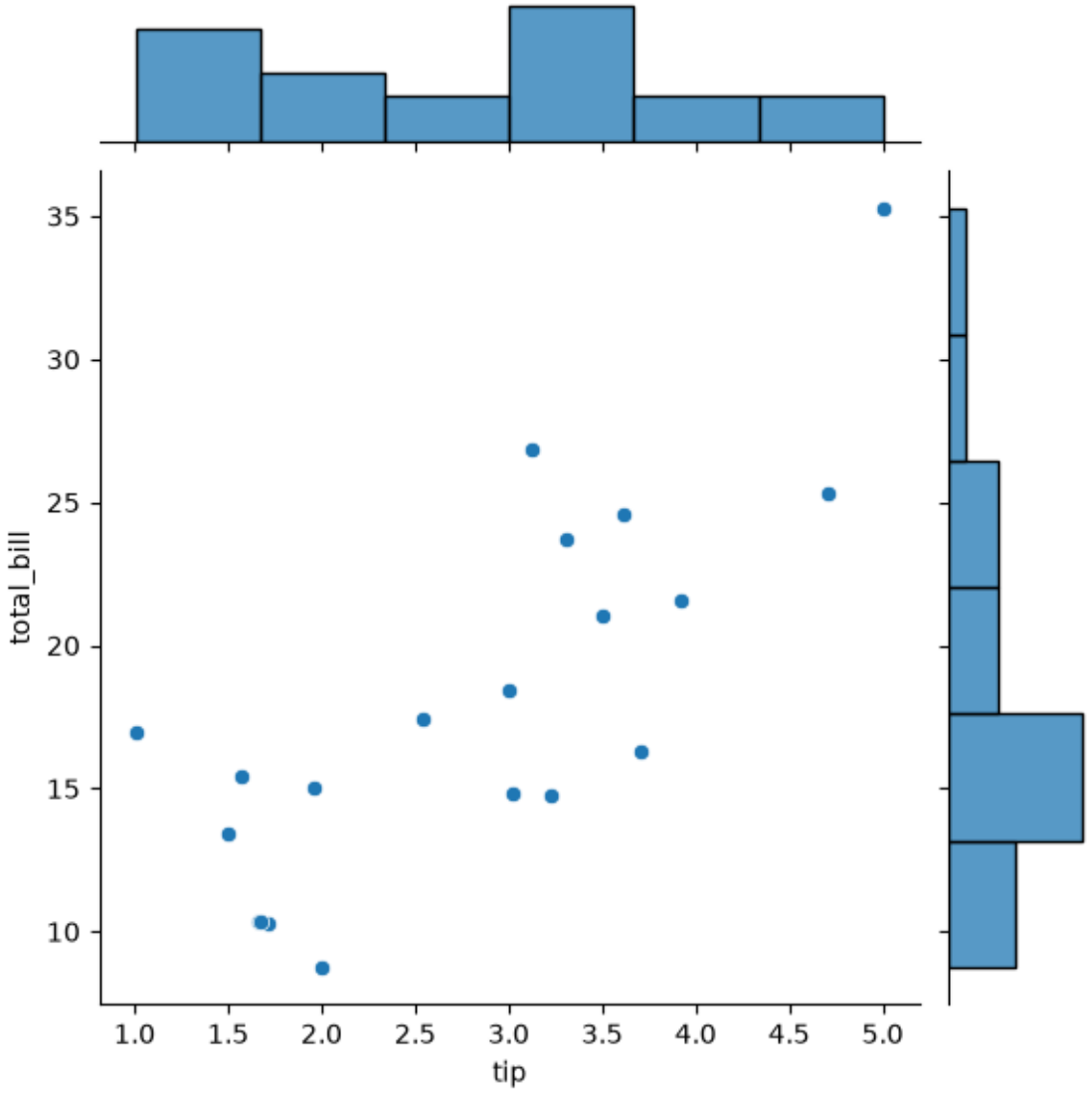
# Bar chart
plt.figure()
tips.sex.value_counts().plot(kind='bar')
plt.show()
```

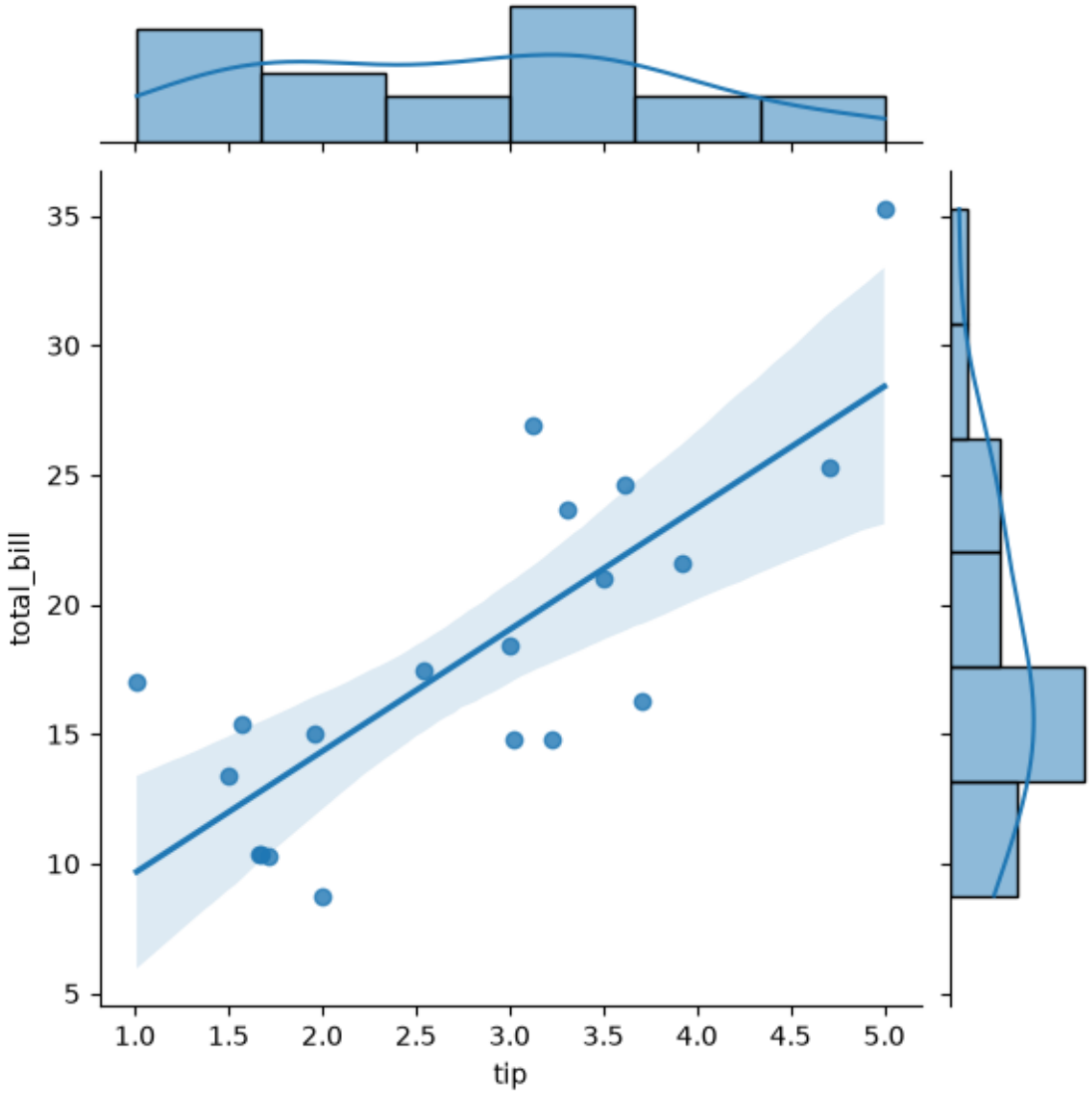
```
# Dinner count by day
plt.figure()
sns.countplot(x=tips[tips.time == 'Dinner']['day'])
plt.show()
```

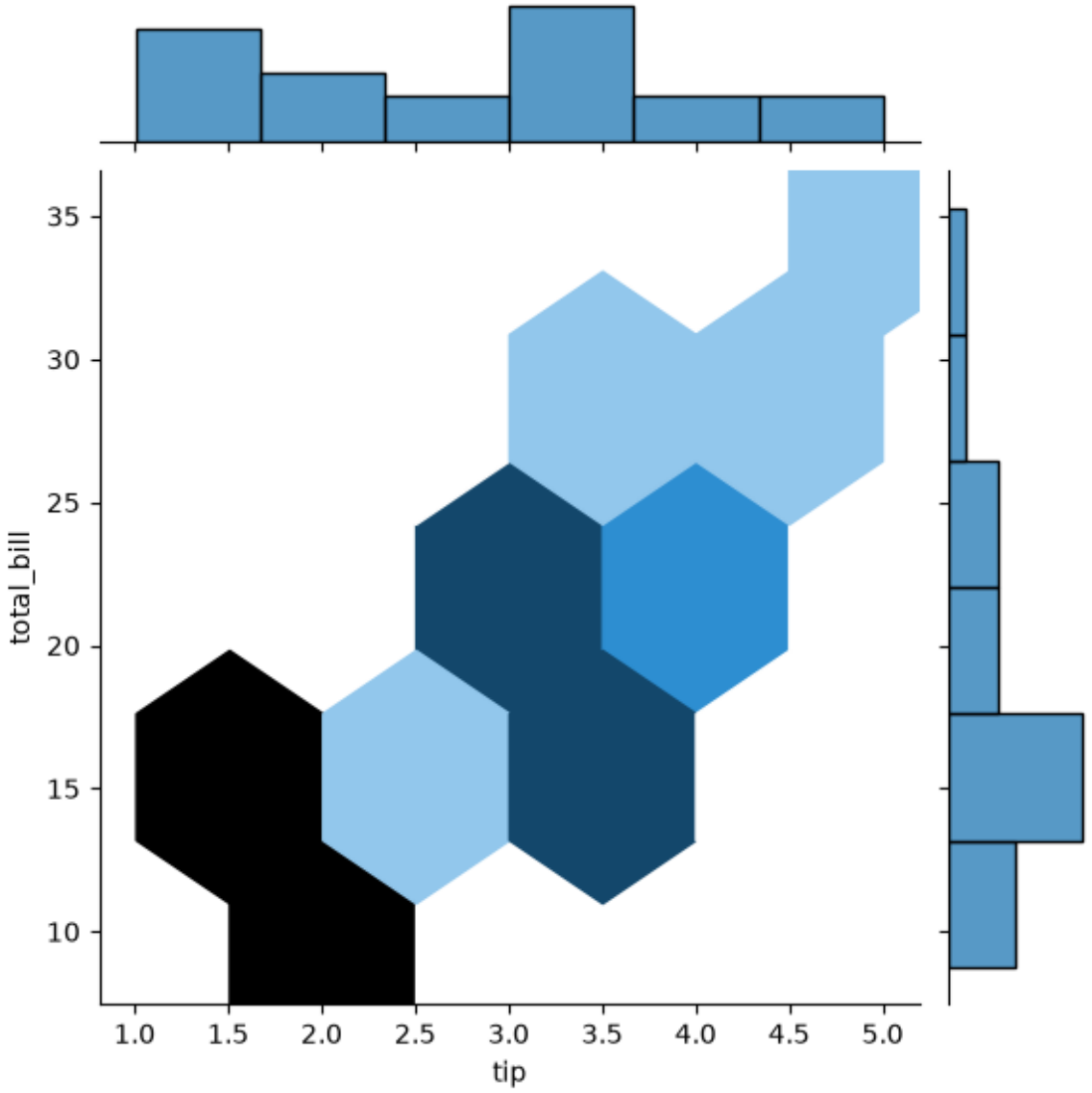
Output:

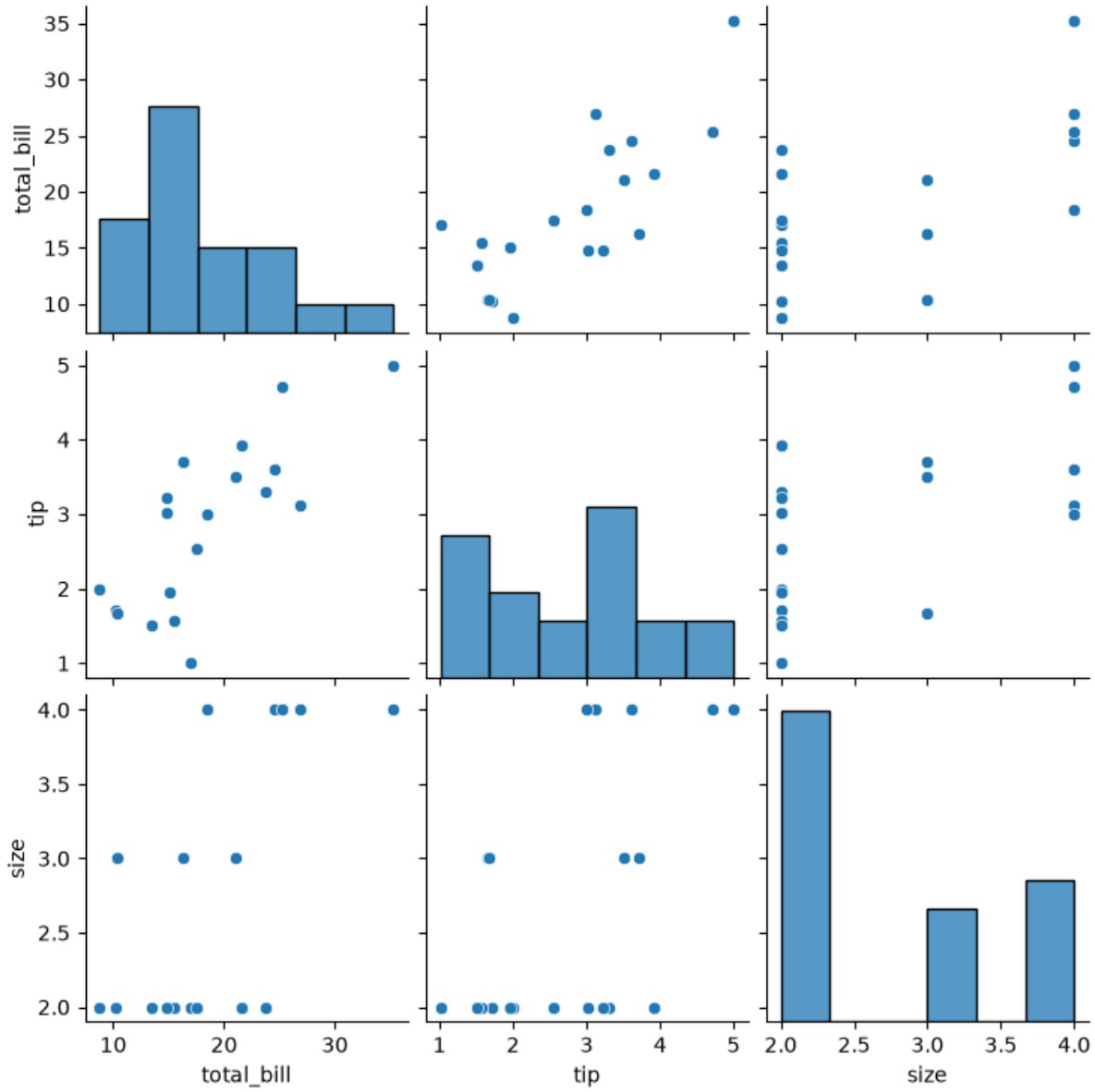


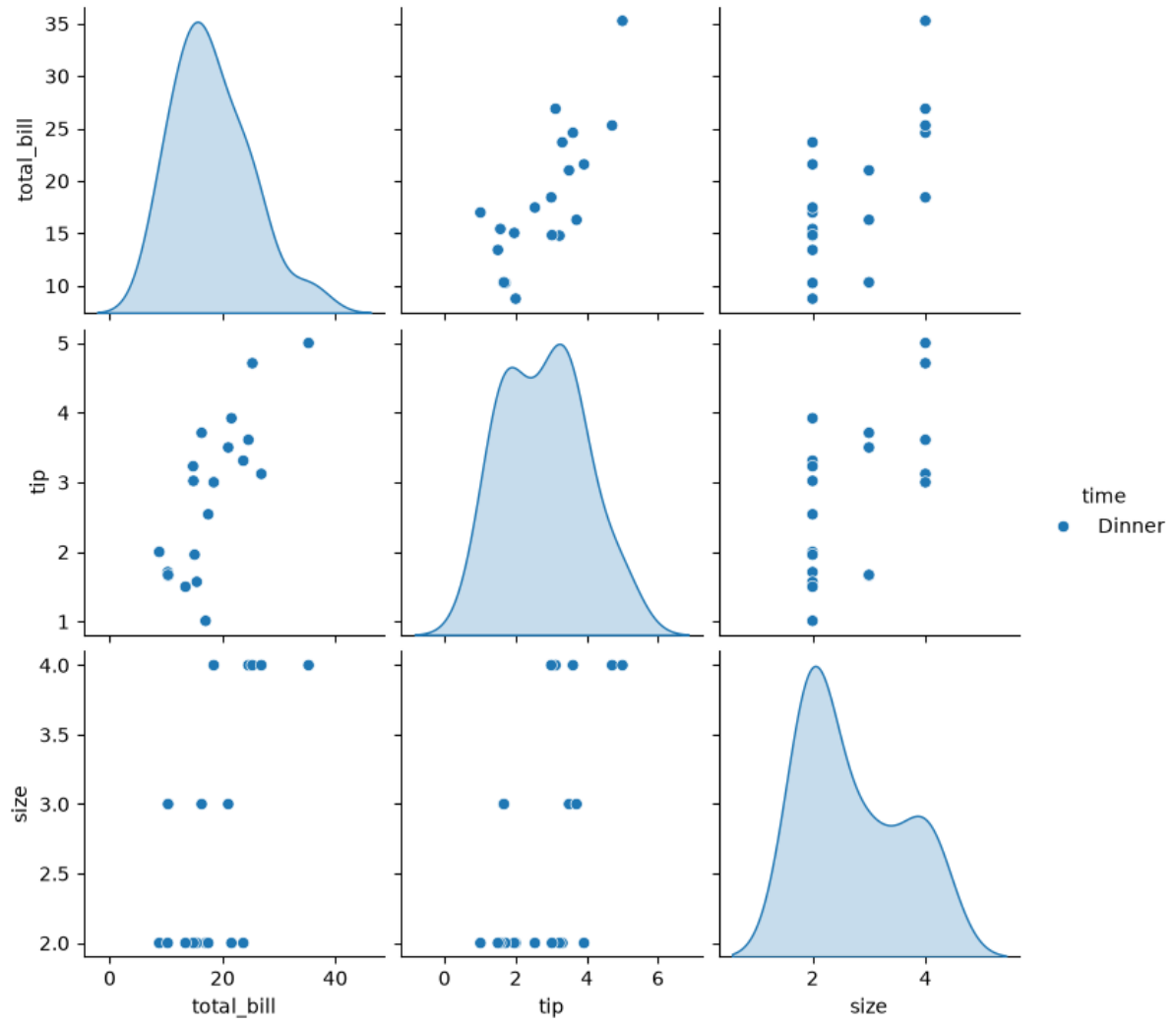

Count
0
1
2
3
4
5
6
7
8
10
15
20
25
30
35
total_bill

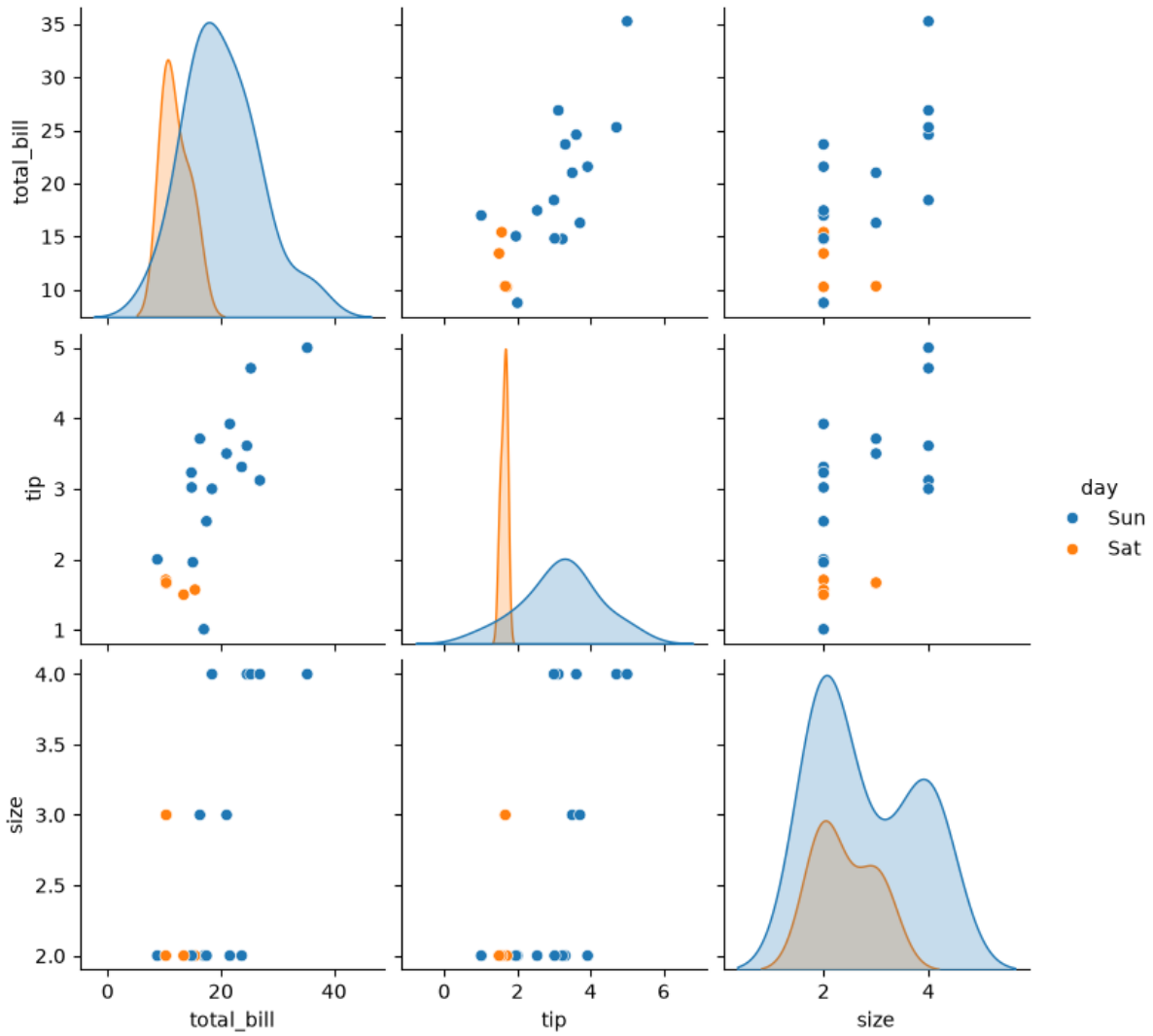

35
30
25
20
15
10
total_bill
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
5.0
tip

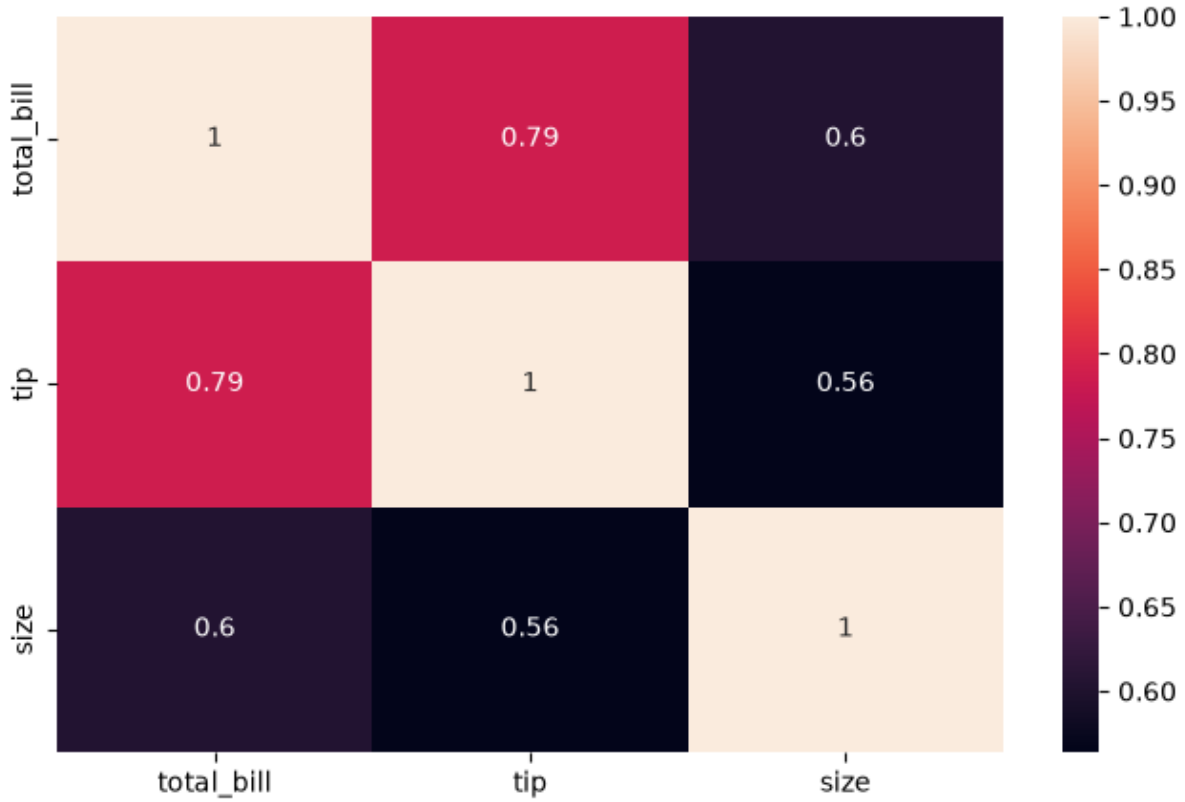

35
30
25
20
15
10
5
total_bill
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
5.0
tip

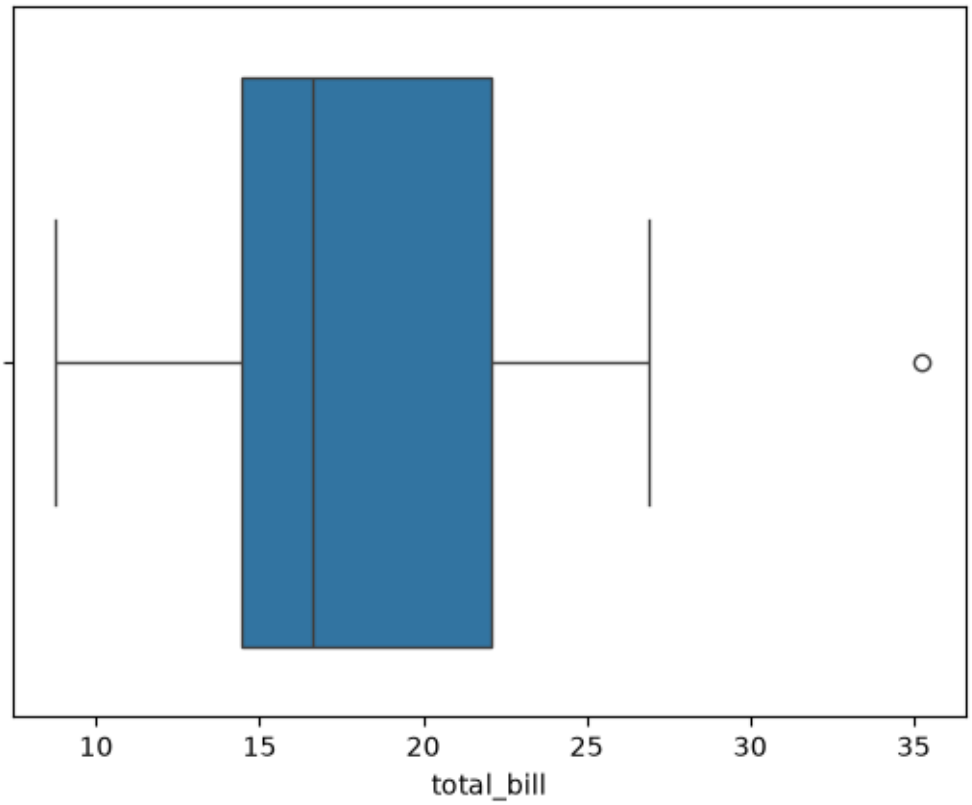

35
30
25
20
15
10
total_bill
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
5.0
tip

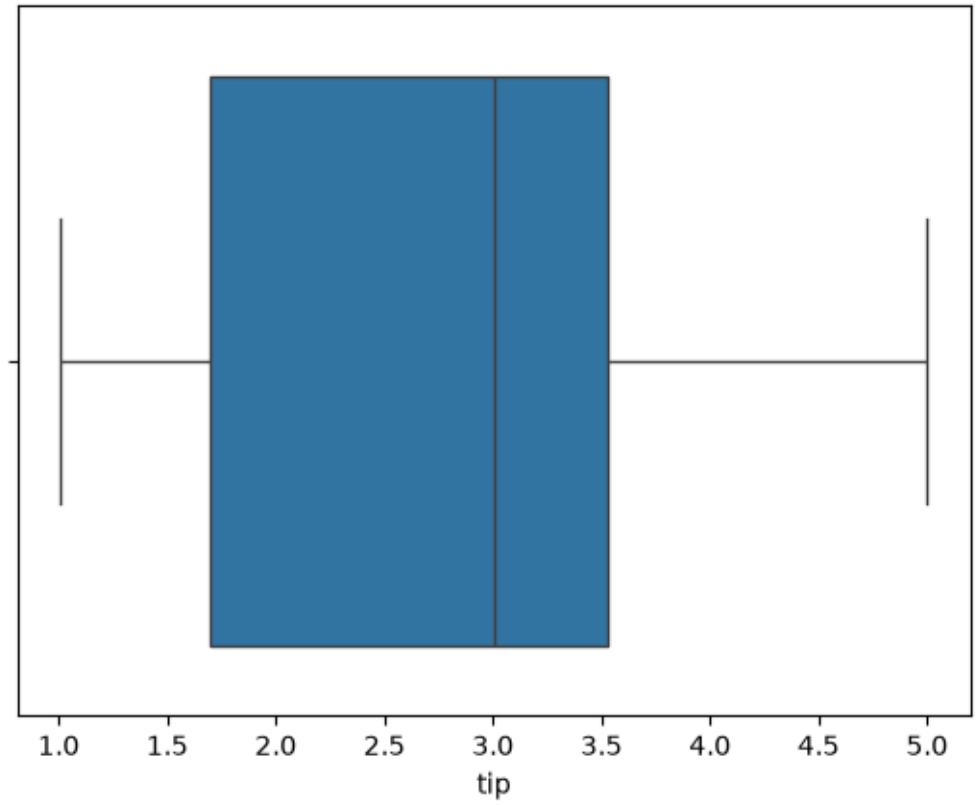

total_bill
tip
size
35
30
25
20
15
10
5
4
3
2
1
4.0
3.5
3.0
2.5
2.0
10
20
30
1
2
3
4
5
2.0
2.5
3.0
3.5
4.0
total_bill
tip
size

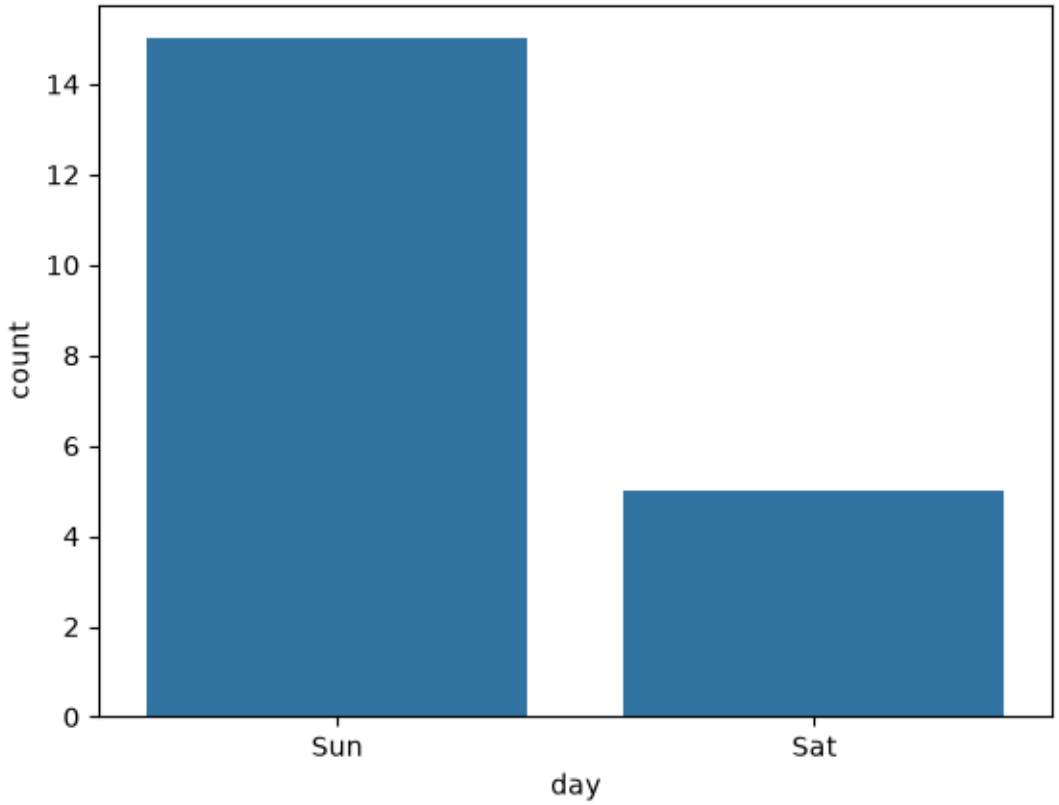

total_bill
tip
size
time
Dinner
0
20
40
0
2
4
6
2
4
10
15
20
25
30
35
1
2
3
4
5
2.0
2.5
3.0
3.5
4.0

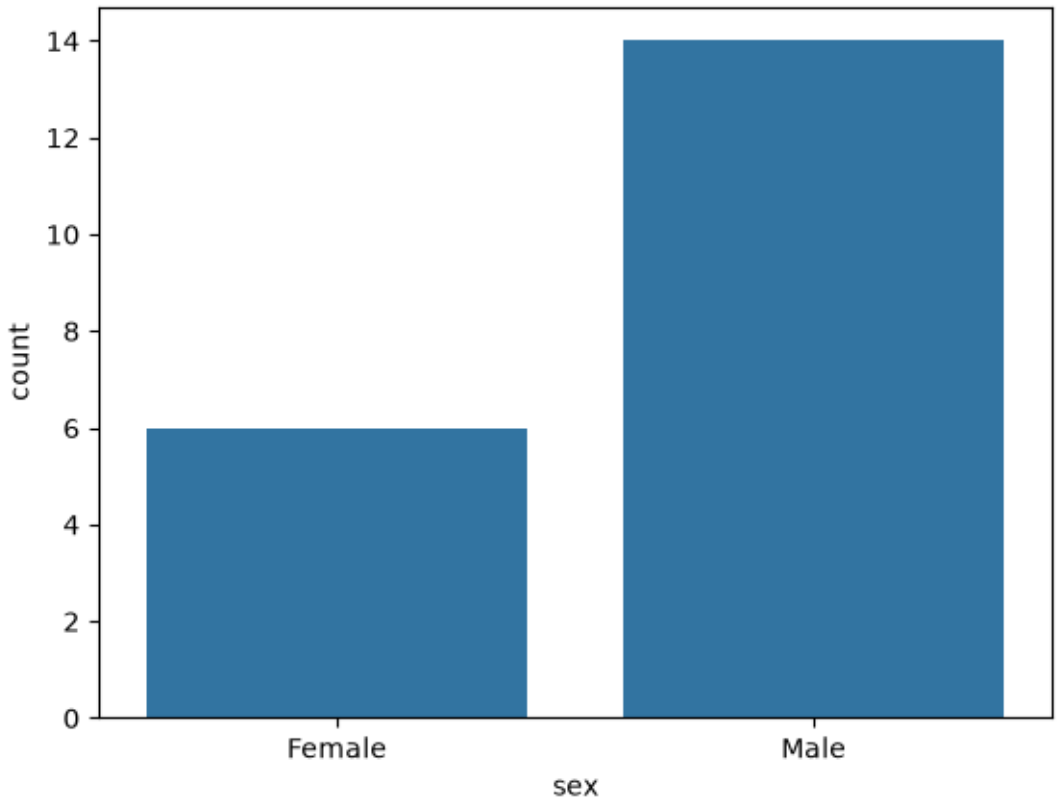

total_bill
tip
size
day
Sun
Sat
35
30
25
20
15
10
5
4
3
2
1
4.0
3.5
3.0
2.5
2.0
0
20
40
0
2
4
6
2
4

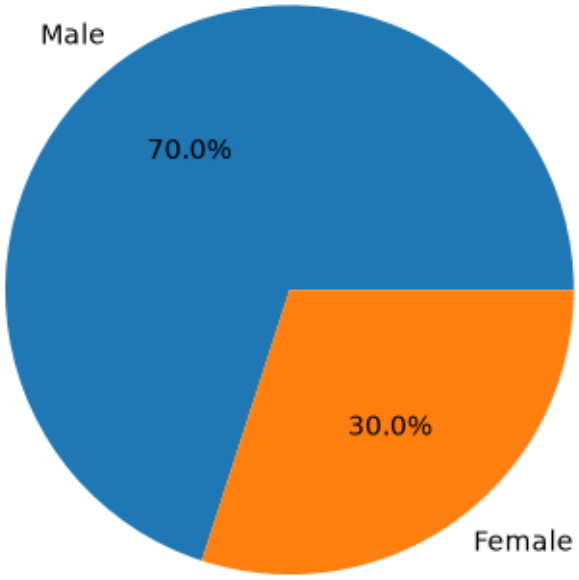

total_bill
tip
size
1
0.79
0.6
0.79
1
0.56
0.6
0.56
1
total_bill
tip
size
1.00
0.95
0.90
0.85
0.80
0.75
0.70
0.65
0.60

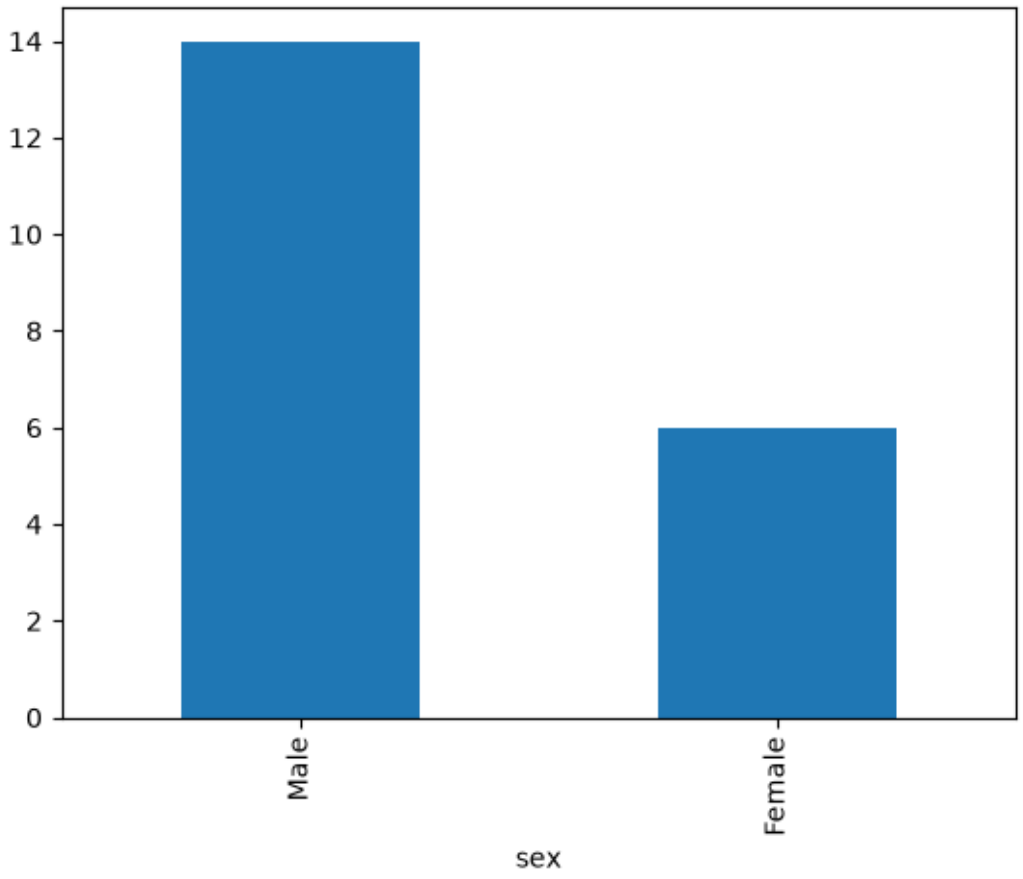

10
15
20
25
30
35
total_bill

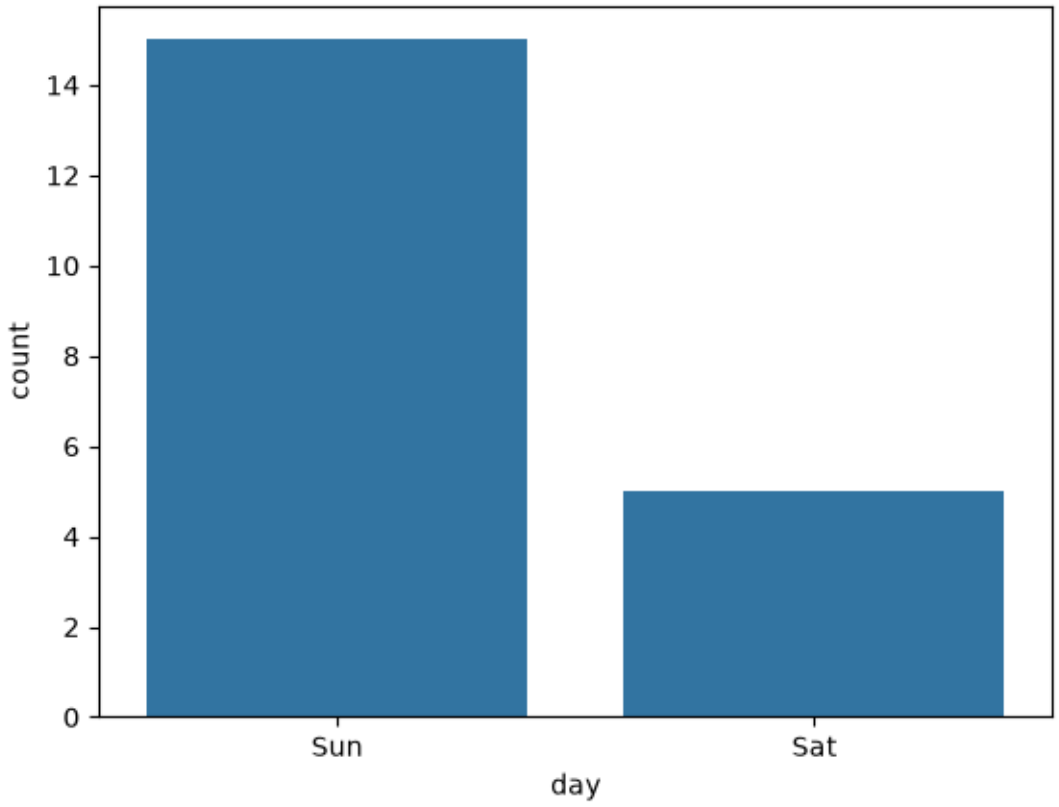

1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
5.0
tip


14
12
10
8
6
4
2
0
count
Sun
Sat
day


14
12
10
8
6
4
2
0
count
Female
Male
sex


Male
70.0%
30.0%
Female


14
12
10
8
6
4
2
0
Male
Female
sex


14
12
10
8
6
4
2
0
count
Sun
Sat
day